CHANDIMA BANDARA

Undergraduate | BSc (Hons) in IT specializing in Data Science

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PROFILE

Data Science undergraduate (BSc Hons IT, SLIT) with hands-on experience building machine learning, computer vision, and NLP systems including RAG pipelines, YOLOv8-based detection models, and transformer fine-tuning alongside full-stack development using Spring Boot and the MERN stack. Comfortable working across the data science and software engineering boundary, from model training to deployment. Seeking a Data Science or Software Engineering internship to contribute and grow as part of a product engineering team.

SKILLS

Languages
Python, Java, JavaScript, SQL

AI / ML & Computer Vision
Machine Learning, TensorFlow, PyTorch, Scikit-learn, YOLOv8, OpenCV, RAG Pipelines, FAISS, LLMs, HuggingFace, NLP / Transformers

Frameworks & Tools
Spring Boot, React Native, Node.js, LangChain, Gradio, Thymeleaf, Expo

Databases & DevOps
MongoDB, MySQL, Git, REST APIs, Postman

PROJECTS

AI-Powered Resume Screening System

- *Python, LangChain, RAG, FAISS, Hugging Face API, Qwen 2.5 7B, Gradio*
- Built a RAG-based recruitment tool that auto-evaluates candidates against job requirements using an LLM (Qwen 2.5 7B via Hugging Face API)
- Automated resume parsing and live portfolio verification through PDF extraction and web scraping, removing the need for manual cross-checking
- Designed an interactive dashboard surfacing match scores, skill gaps, and hiring recommendations for recruiters
- Links: GitHub Repository | Live Demo

AI Car Damage Detection & Verification System

- *Python, YOLOv8s, ResNet-18, OpenCV, Gradio*
- Built a computer vision pipeline that automates vehicle inspection by comparing before/after photos and detecting newly acquired damage
- Implemented vehicle identity verification using ResNet-18 feature embeddings and cosine similarity to prevent spoofing
- Applied ORB keypoint detection and homography-based perspective alignment to correct for camera angle and movement between image pairs
- Trained a custom YOLOv8s model over 50 epochs for damage detection across 7 vehicle panel classes, using IoU filtering to isolate genuinely new damage
- Reduced false positives from background clutter by augmenting the training dataset with null/empty-environment images
- Deployed the system as an interactive Gradio app on Hugging Face Spaces
- **Links:** GitHub Repository | Live Demo

Student Mental Health Early Detection System

- *Python, Scikit-learn, Keras/TensorFlow, MLP, EDA*
- Developed an end-to-end ML pipeline to detect early signs of mental health risk in students based on academic and stress-related factors
- Led data preprocessing — cleaning, transformation, and outlier removal — on a Kaggle dataset
- Designed a comparative modeling pipeline evaluating KNN, SVM, Logistic Regression, Decision Trees, and Random Forest
- Trained and optimized an MLP neural network, achieving 90.91% peak accuracy

- Generated evaluation metrics (confusion matrix, accuracy trends, loss curves) to validate model performance
- **Links:** GitHub Repository | Live demo

Student Concern Management System (SCMS)
- *Java 21, Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, MS SQL Server, Google Gemini API*
- Collaborated with a real client to build a role-based concern management system for Students, Admins, and Owners
- Implemented secure authentication — email OTP, password recovery, BCrypt encryption — with role-based access control across all user types
- Built workflow-driven concern tracking with evidence uploads, status management, and meeting scheduling
- Integrated the Google Gemini API for AI-powered moderation of community forum content
- Developed a centralized notification system supporting both direct and broadcast communication
- **Links:** GitHub Repository | Live Demo | Project Details

Student Concern Management System (SCMS) – Mobile App
- React Native, Expo, Node.js, Express.js, MongoDB (Atlas), Mongoose, JWT
- Developed a full-stack, multi-role mobile app (MERN stack) to streamline communication between students and academic administration
- Architected dedicated dashboards and workflows for Student, Admin/Consultant, and Owner roles using a clean MVC pattern
- Built a real-time concern tracking pipeline with status updates and automated email notifications
- Developed a secure medical triage module with file uploads and restricted access for sensitive health data
- Added institutional oversight features including performance analytics, PDF/Excel report generation, and broadcast announcements
- **Links:** GitHub Repository | Project more details

VeloRent - Web-Based Car Rental System
- *Java, Spring Boot, MS SQL Server, Thymeleaf, RESTful APIs*
- Developed a web application to automate the vehicle rental lifecycle with real-time booking and secure authentication
- Served as Lead Developer for security, implementing authentication, two-factor authentication, and password recovery
- Implemented role-based access control across Customers, Owners, Fleet Managers, and Admin roles
- Designed backend logic and dashboards for Fleet Managers to track vehicle availability, active rentals, and returns
- Collaborated on integrating payment processing, inventory management, and dynamic pricing modules
- **Links:** GitHub Repository

EDUCATION

BSc (Hons) in Information Technology, 2024 – present
Sri Lanka Institute of Information Technology (SLIT) - Malabe, Sri Lanka

Advanced Level (AL) — Technology Stream 2022
Sir John Kotelawala C.C. - Dodamgaslanda, Sri Lanka

Ordinary Level (OL) 2019
Namal Anga M.V. - Kurunegala, Sri Lanka